Supplementary Tables for Submission 991

Full supporting tables (Set A)

This document gives the full versions of three tables summarized in our OpenReview response; the response is self-contained and this material is supporting only. All numbers use fresh rebuttal-period seeds and may differ from the submitted paper by up to about 0.04 on coverage and width, within seed variation. "Paper" denotes the submitted internal-split protocol (values quoted from the paper); "Strict" denotes the external 40/10/50 split.

Note on the choice of split conformal (R1)

Reviewer YU66 also named cross-conformal and leave-one-out as alternatives. We use split conformal for its clean exchangeability guarantee and its tractability at our scale (3 judges, 2 benchmarks, 8 conformal methods, 10 seeds per cell). Cross-conformal and jackknife+ are valid but substantially costlier in retraining, and jackknife+ attains only $1 - 2\alpha$ marginal coverage (Barber et al., 2021), which would weaken rather than strengthen the reported guarantees. Table A1 already confirms valid coverage for all three fitted-model methods the review named, under a proper training/calibration split, so the validity concern is addressed without these alternatives; we note them as future work.

Table A1. All fitted-model methods under the strict external split (R1)

R2CCP, CQR, and CHR under the submitted protocol and the strict external 40/10/50 split (3 judges, 10 seeds per strict-split cell; mean ± s.d.). Δ columns are paired t-test p-values on matched rebuttal-period runs. R2CCP coverage holds at 0.90 under both protocols; CQR over-covers throughout (weak quantile-regression features); CHR recovers nominal coverage under the strict split once the finite-sample $(1+1/n)(1-\alpha)$ correction is included (standard split-CHR, Sesia & Romano, 2021).

Method	Judge	Paper		Strict 40/10/50		Paired p	
		cov	w	cov	w	Δcov	Δw
R2CCP	LLaVA-Critic	0.900	3.05	0.894 ± 0.015	3.041 ± 0.095	0.772	0.093
R2CCP	Phi-4	0.891	3.13	0.890 ± 0.012	3.145 ± 0.076	0.377	0.908
R2CCP	Gemini	0.898	2.85	0.897 ± 0.012	2.887 ± 0.067	0.566	0.015
CQR	LLaVA-Critic	0.996	3.95	0.970 ± 0.011	3.60	0.002	0.002
CQR	Phi-4	0.997	3.98	0.993 ± 0.009	3.91	0.641	0.469
CQR	Gemini	0.996	3.95	0.977 ± 0.012	3.65	0.002	0.002
CHR	LLaVA-Critic	0.880	2.97	$\mathbf{0.900 \pm 0.014}$	3.29	0.002	0.002
CHR	Phi-4	0.896	3.16	$\mathbf{0.902 \pm 0.015}$	3.33	0.004	0.002
CHR	Gemini	0.880	2.80	$\mathbf{0.906 \pm 0.012}$	3.20	0.002	0.002

Table A2. Per-judge risk-coverage AURC (R2)

AURC (lower is better) per judge, with the weighted-average and the pooled single-threshold rows. CP width beats random within every judge. Judge-internal confidence (margin, entropy) is informative within each judge (weighted average better than random) but collapses to random when a single global threshold is pooled across judges, because logprob confidence is not comparable across models. CP width, expressed in shared score units, is essentially unchanged between the weighted-average and pooled rows, so a global threshold transfers across judges.

Judge (n)	Random	CP width	Judge margin	Judge entropy
Gemini (27,037)	1.122	**0.948**	1.080	1.129
Phi-4 (28,274)	1.001	**0.926**	0.991	0.995
LLaVA-Critic (28,583)	1.032	0.993	**0.948**	**0.948**
Weighted avg over judges	1.051	0.956	1.005	1.022
Pooled (single threshold)	1.053	0.963	1.050	1.069

Table A3. Full width-decile reliability on MLLM-as-a-Judge (R2)

Per-instance triples pooled across 3 judges and 10 seeds (n = 83,894 after dropping unparseable outputs), binned into deciles by raw CP width. Accuracy falls near-monotonically as width grows (one small inversion at deciles 5 to 7): the tightest decile reaches 43.9% exact accuracy and the widest 25.5%, a 1.7× separation from the CP-width ordering alone.

Width decile (raw)	mean width	exact acc.	±1 acc.	judge MAE
$(0.35, 2.30]$	2.12	43.9%	81.9%	0.84
$(2.30, 2.53]$	2.42	37.9%	77.1%	0.97
$(2.53, 2.71]$	2.62	35.1%	75.3%	1.01
$(2.71, 2.90]$	2.80	33.5%	76.3%	1.01
$(2.90, 3.05]$	2.99	31.9%	74.2%	1.06
$(3.05, 3.14]$	3.09	32.9%	75.2%	1.03
$(3.14, 3.25]$	3.20	31.9%	73.9%	1.05
$(3.25, 3.41]$	3.33	30.2%	71.5%	1.10
$(3.41, 3.68]$	3.52	26.9%	67.3%	1.21
$(3.68, 4.00]$	3.93	25.5%	65.3%	1.25